

DAF26BX06-NV510: AI/ML for Next Generation of Missile Detection, Warning, Tracking, and Reporting

PROJECTED CMMC LEVEL REQUIREMENT:

Level 2 (Self)

TECHNOLOGY AREAS:

Space Platforms

MODERNIZATION PRIORITIES:

Space Technology

KEYWORDS:

SpaceWERX, Space Platforms, Space Sensing, Overhead Persistent Infrared (OPIR), Data Fusion, AI/ML, OODA, Mission Assurance

ADDITIONAL INFORMATION

N/A

OBJECTIVE:

SpaceWERX, in partnership with Space Systems Command's Space Sensing System Delta 84 and the OPIR Tap Lab, are seeking innovative solutions supporting the next generation of missile detection, warning, tracking, and reporting. The objective of this SBIR Phase I effort is to evaluate the feasibility of applying innovative data processing techniques including the use of Artificial Intelligence and Machine Learning (AI/ML) methodologies to address the evolving challenges in missile detection, warning, tracking, and reporting. The primary goal of this research and development initiative is to advance data fusion, sensor integration, and real-time processing methods that enhance the accuracy, responsiveness, and reliability of satellite-based missile warning systems. This includes addressing the data interoperability from multi-orbit (GEO, MEO, LEO) satellite constellations, optimizing on-orbit and ground-based processing frameworks, and improving data quality from diverse sensing modalities from legacy and future Overhead Persistent Infrared (OPIR) space architectures. Specifically, this effort will focus on researching and developing state-of-the-art algorithms and processing techniques that support rapid and precise multi-sensor fusion, robust target identification, and accurate real-time 3D tracking of traditional, hypersonic, and maneuverable missile threats. By leveraging novel techniques in data association, sensor fusion, and edge processing, the solution will aim to mitigate limitations presented by diverse sensor resolutions, data cadences, and observational geometries. The project will also investigate methods to increase computational efficiency for onboard satellite processing and consider robustness against noise in highly dynamic operational environments.

ITAR:

The technology within this topic is restricted under the International Traffic in Arms Regulation (ITAR), 22 CFR Parts 120-130, which controls the export and import of defense-related material and services, including export of sensitive technical data, or the Export Administration Regulation (EAR), 15 CFR Parts 730-774, which controls dual use items. Offerors must disclose any proposed use of foreign nationals (FNs), their country(ies) of origin, the type of visa or work permit possessed, and the statement of work (SOW) tasks intended for accomplishment by the FN(s) in accordance with section 3.5 of the Announcement. Offerors are advised foreign nationals proposed to perform on this topic may be restricted due to the technical data under US Export Control Laws.

DESCRIPTION:

The purpose of this topic is to address a critical need within the U.S. Space Force's Missile Warning, Tracking, and Response mission by advancing the application of artificial intelligence (AI) and machine learning (ML) technologies to improve multi-sensor fusion, missile tracking precision, and real-time situational awareness within a multi-orbit satellite framework (GEO, MEO, and LEO). Current technologies face challenges in fusing large volumes of disparate data across multiple sensing modalities, tackling latency in data processing, and accurately

tracking increasingly sophisticated threats such as hypersonic and maneuverable missile systems.

This effort seeks to develop and validate advanced data processing techniques including AI/ML solutions for both ground-based and satellite-based multisensor fusion capable of real-time processing for advanced OPIR Missile Detection, Warning, Tracking, and Reporting. These solutions will enable enhanced detection, tracking, and reporting of advanced missile threats by leveraging novel algorithms for target identification, data association, and real-time 3D tracking across multi-orbit platforms. Initial solutions should target feasibility studies and technical proof-of-concept demonstrations with the goal to transition the capability into the OPIR Tap Lab or similar testbed to build and test solutions in an operationally relevant environment. The advancements made on this effort have the potential to significantly contribute to the Space Force's mission of persistent global surveillance, improved defensive decision-making, and sustained space superiority.

PHASE I:

Phase I will assess the feasibility of AI/ML-driven multi-sensor fusion techniques for real-time advanced missile detection, warning, and tracking across multi-orbit satellite systems and data architectures. The desired future state of this effort is a network-aware, autonomous constellation (spanning multiple orbital regimes) where ground "migrates" with space via edge fusion that compresses timelines for joint fight without loss of quality or confidence.

The goal of this Phase I effort is to onboard and coordinate emergent solutions into the OPIR TAP lab and create a shared understanding of the problem set of the Missile Warning and Tracking (MWT) mission area. Successful Phase I efforts from this topic will target capability testing within the OPIR Tap Lab with the goal of advancing capability and de-risking follow-on capability development in subsequent phases. Proposed solutions should include a feasibility study and algorithm prototypes or simulation results as applicable as well as a clear path to integration and testing within the OPIR Tap Lab.

Focused topic areas can include any part of the following:

- Edge Fusion & Autonomy: AI/ML solutions for on-orbit track fusion, autonomous constellation tasking, and event-driven scheduling across proliferated MEO/LEO architectures in contested environments.
- Multi-Source Integration: Capabilities that fuse OPIR, EO, radar, SIGINT, and commercial sensor data to improve detection, tracking, typing, and characterization — with scalable software architectures and ML-driven sensor calibration.
- Observe, Orient, Decide, Act (OODA) Loop Collapse & Mission Assurance: Solutions that accelerate the sensor-to-decision-maker kill chain with sufficient confidence and explainability to assure mission-critical decisions. These can include but are not limited to dynamic user control of automation (C2, operator overload, scalable operations), orchestration, edge computing, AI trade-space exploration, and authoritative knowledge access.
- New sources and methods for Missile Warning that are outside of the current MW operational paradigm tactics, techniques, and procedures (TTPs).

Proposals should emphasize technical feasibility and clearly outline a Phase I research plan that can include modeling, simulation, literature review, or trade studies. While a Phase II transition concept may be outlined, Phase I deliverables are expected to focus on a feasibility study detailing USSF CONOPS and technical viability, as well as analytical findings.

Feasibility Study: A feasibility study is an assessment of the practicality of developing and implementing a new technology within a given concept. The study aims to objectively and rationally uncover the strengths and weaknesses of the new technology, the feasibility of successful development of the technology, and the viability of the success of implementation of the technology in the proposed CONOPS. The study should:

- Identify the benefits of the new technology within the CONOPS over existing solutions (or lack thereof)
- Identify the risks, uncertainties/unknowns, and issues of the proposed CONOPS with mitigation and closure measures
- Identify the critical technology elements (CTEs) of the new technology
- Provide data to support the feasibility of technology development of each CTE to include: literature research supporting technical viability or historical usage, trade studies identifying existing technologies, current developments, and/or feasible development pathways, analysis demonstrating technical solutions, and heritage usage of similar technology with identified differences in operations and/or environment for proposed CONOPS.

The Concept of Operations (CONOPS) should clearly communicate how the technology is intended to support a military mission within a defined system. The main purpose of the CONOPS is to facilitate a common understanding of a future system to help develop operational and system-level requirements. CONOPS should be written at the system level first and then expanded on how the technology functions enable successful military operations.

PHASE II:

Phase II aims to build off the work completed from the Phase I after successfully demonstrating capability for operational use within the OPIR Tap Lab or similar environments that demonstrate integrating advanced AI/ML for real-time multi-sensor fusion and missile tracking from OPIR systems. Activities include system development, integration, and testing in operationally relevant environments to verify performance metrics (e.g., accuracy,

latency, resilience, coverage). Phase II will mature the feasible concepts identified in Phase I into a deployable, testable prototype. Phase II will include design refinement, system development, and operationally relevant testing focused on delivering a measurable improvement in AI/ML techniques to detect and/or accurately fuse any source detecting these threats, either on the ground in the traditional processing ground stations or on board via edge processing.

While successful Phase I efforts establish technical feasibility and transition the validated proof-of-concept into the OPIR Tap Lab or a similar testbed, the Phase II efforts will continue the development of technologies coming out of the OPIR Tap Lab. By utilizing this operationally relevant environment, the work progresses to system development, integration, and testing, which de-risks follow-on capability development. The ultimate goal of this pathway is to mature the feasible concepts into a deployable, testable working prototype that delivers a measurable improvement in techniques for multi-sensor fusion and missile detection, warning, tracking and reporting.

PHASE III DUAL USE APPLICATIONS:

Potential Phase III efforts will pursue transition of the capability to operational use through non-SBIR/STTR funding streams, aiming for a Technology Readiness Level (TRL) of 6 or 7 and demonstrating functionality in relevant environments. A key transition pipeline for this capability is the Space Modernization Initiative (SMI) program within Space System Command's Space Sensing portfolio. Successful efforts in potential Phase III efforts will involve full integration into OPIR operational data workflows and fielding the capability at scale to provide recurring operational value and reduce warfighter risk by expanding OPIR sensing capabilities.

REFERENCES:

1. TRL Guide - <https://www.gao.gov/assets/gao-20-48g.pdf>.
2. OPIR Tap Lab - <https://boulderlab.org/>.

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